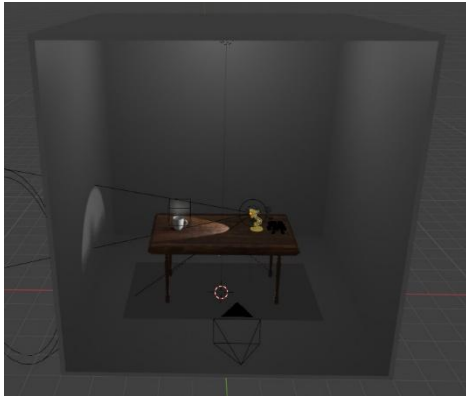


Project for the course „3D Modelling and Animation“ - documentation

Dino Gabrić

In this project, an animation of a lamp was created: it wakes up and looks around the room before finally looking into the camera. The inspiration for this work was the Pixar lamp.

The scene consists of a room (made from a cube mesh) and one face of the cube deleted, through which the camera looks. A dark material/color was applied to the room so that the lamp stands out and remains the center of attention.



The lamp model was taken from the Blender extension BlenderKit, and its material was changed so it would stand out more in the dark environment. In addition, a custom-made skeleton (i.e., an armature) was added to make its movement and keyframe animation easier.



The table on which the lamp and the other parts of the scene are placed was made using polygonal modeling and was given a wood texture. The table legs were also created with polygonal modeling and assigned the same texture.



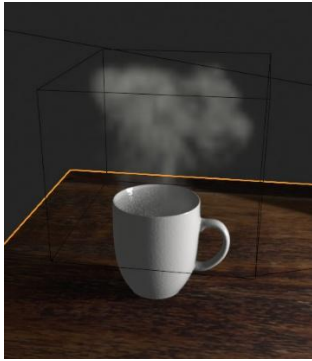
There is also a glass on the table, modeled using a Bézier curve.



Besides the glass, there is a spider model on the table, taken from <https://free3d.com/3d-model/spider-257546.html>, to which a hair/fur particle system was added with 5000 strands, and its color was changed.



There is also a coffee cup on the table; its model was taken from the BlenderKit extension. A smoke/steam effect was added to it (to create the illusion of hot coffee) by inserting a cylinder (disk) above the model, from which the effect “comes out” using a physics engine and a particle system, creating the impression of a hot liquid in the cup.



The scene contains two light sources: one low-power light (a point light) to lightly illuminate the scene, and the other connected to the sphere (“bulb”) inside the lamp. This second light is a spotlight, used to draw the viewer’s attention depending on where the lamp is “looking.”

The lamp was animated using keyframes, and in the Graph Editor various interpolation curves were used (Bézier, constant, linear, exponential) to achieve the most realistic possible look of the lamp’s movement through space—that is, within the room.